ATAI-8WD

Document-000 Master Engineering Specification

Ultra Safe Capsule Hybrid Airship VTOL Platform

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1. Executive Overview

ATAI-8WD is a hybrid VTOL airship platform combining buoyant lift from helium with high-power ducted-fan propulsion and a modular safety architecture. The concept is designed for heavy-lift logistics, industrial operations, and safe crew transport.

The architecture uses three functional layers:

- Layer-1: Propulsion Deck primary lift and propulsion
- Layer-2: Helium Lift Platform buoyant lift and stability
- Layer-3: Ultra Safe Capsule crew cockpit and autonomous survival module

The design philosophy is redundancy, modular separation, and extreme operational safety.

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2. Tri-Layer System Architecture

Layer-1   Propulsion Deck

Dimensions:
50 m Ù 50 m Ù 10 m

Purpose:
- Main powered lift
- Propulsion generation
- Energy generation and distribution

Primary propulsion components:
- 4 Ù 20 m ducted lift fans (vertical lift)
- 8 Ù 10 m vectored fans (45° downward)

Estimated operational mass:
143 tons

Layer-2   Helium Lift Platform

Dimensions:
70 m Ù 70 m Ù 30 m

Purpose:
- Provide static buoyant lift
- Reduce propulsion power requirements
- Increase hover endurance

Helium volume:
115,000   147,000 m^3

Net buoyant lift:
≈ 120 tons

Estimated structural mass:
≈ 58 tons

Layer-3 Ultra Safe Capsule

Dimensions:
30 m Ù30 m Ù10 m

Purpose:
- Crew cockpit
- Flight control center
- Emergency survival module

Propulsion:
- 9 Ù9.63 m ducted lift fans
- 8 Ù5 m control fans at 45°

Estimated operational mass:
≈ 50 tons

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3. Helium Lift Physics

Air density:
1.225 kg/m^3

Helium density:
0.178 kg/m^3

Net buoyant lift:
≈ 1.047 kg per m^3

For approximately 115,000 120,000 m^3 of helium:

Total buoyant lift ≈ 120 tons

This reduces the effective powered lift requirement of the vehicle.

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4. Layer-1 Propulsion System

Primary Lift Fans

Quantity: 4
Diameter: 20 m
Disk area per fan: 314 m²

Estimated operational lift per fan:
$\approx 70 \quad 75$ tons

Total primary lift capability:
≈ 300 tons

Secondary Control Fans

Quantity: 8
Diameter: 10 m
Angle: 45° downward

Vertical lift component per fan:
≈ 13 tons

Total additional vertical lift:
≈ 104 tons

Total Propulsion Lift Capacity

Primary lift: ~300 tons
Secondary lift: ~104 tons

Maximum total thrust:
≈ 400 tons

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5. Capsule Propulsion System (Layer-3)

Main Capsule Lift Fans

Quantity: 9
Diameter: 9.63 m

Estimated lift per fan:
$\approx 9 \quad 10$ tons

Total lift:
≈ 90 tons

Capsule Control Fans

Quantity: 8
Diameter: 5 m
Angle: 45°

Vertical lift per fan:
≈ 2.5 tons

Total additional lift:
≈ 20 tons

Total Capsule Lift

Approximately 110 tons of lift.

This allows the capsule to operate independently during emergency separation.

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6. Engine System

Layer-1 Powerplant

Type: Industrial gas turbine

Total mechanical power:
≈ 60 MW

Functions:
- Drive large lift fans
- Generate electrical power
- Support propulsion and control systems

Capsule Powerplant

Type: Hybrid turboshaft system

Power output:
≈ 12 MW

Functions:
- Independent VTOL capability
- Emergency flight operations

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7. Fuel Systems

Layer-1 Fuel System

Tank volume:
≈ 71 m^3

Fuel type:
Jet-A / Sustainable Aviation Fuel

Fuel mass:
≈ 57 tons

Capsule Fuel System

Tank volume:
≈ 12 m³

Fuel mass:
≈ 10 tons

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8. Weight Breakdown

Layer-1 Propulsion Deck

- Structure: 40 tons
- Engines: 20 tons
- Fans: 15 tons
- Fuel: 57 tons
- Systems: 11 tons

Total:
≈ 143 tons

Layer-2 Helium Platform

- Structure: 58 tons
- Helium lift capability: 120 tons

Layer-3 Ultra Safe Capsule

- Structure: 18 tons

- Fans: 12 tons
- Engines: 10 tons
- Fuel: 10 tons

Total:
≈ 50 tons

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9. Total Vehicle Mass

- Layer-1: 143 tons
- Layer-2: 58 tons
- Layer-3: 50 tons

Total system mass:
≈ 251 tons

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10. Effective Lift Requirement

Helium lift: 120 tons
Total weight: 251 tons

Effective powered lift requirement:
≈ 131 tons

Available propulsion lift:
≈ 400 tons

Thrust-to-weight margin:
$\approx 3 : 1$

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11. Flight Performance

Hover endurance (full system):
≈ 3.5 hours

Estimated cruise range:
$\approx 500 \quad 550$ km

Cruise speed:
$\approx 140 \quad 180$ km/h

Capsule independent hover time:
≈ 1.8 hours

Capsule range:
≈ 230 km

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12. Safety Systems

Global parachute system:
16 parachutes protecting the entire vehicle.

Capsule parachute system:
16 parachutes dedicated to the Ultra Safe Capsule.

Landing gear:
12 hydraulic landing legs for hard-surface or emergency landings.

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13. Layer Separation Strategy

Stage 1:
Separation of propulsion deck if structural failure occurs.

Stage 2:
Helium platform release to reduce mass.

Stage 3:
Autonomous capsule flight and controlled landing.

This layered separation dramatically increases crew survivability.

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14. Industrial Applications

The ATAI-8WD platform enables multiple new aerial industrial roles.

- Heavy lift aerial crane
- Wind turbine installation
- Transport of large industrial equipment
- Oil and gas logistics
- Mining infrastructure transport
- Disaster relief logistics
- Remote construction support
- Military heavy logistics

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15. Scalable Architecture

The concept can scale to larger platforms.

Example models:

ATAI-8WD
Helium lift: 120 tons
Total thrust: ~400 tons

ATAI-20HD
Helium lift: ~300 tons
Total thrust: ~800 tons

ATAI-50HD
Helium lift: ~700 tons
Total thrust: ~1500 tons

Large variants could function as true aerial construction cranes.

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16. Strategic Impact

Hybrid airship-VTOL systems like ATAI-8WD could create a new class of industrial aircraft capable of transporting extremely heavy loads without requiring runways or major infrastructure.

The system combines buoyant lift efficiency with powered VTOL control and modular safety architecture.

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